

IOT-2043
Examination –May- 2024
B.Tech. IV Sem : Internet of Things
Python programming

Max. Marks : 70

Min. Marks : 22

and number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a – Max 50 words, Part b – Max 50 words,

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

Marks	BL	CO
02	1	1
02	1	1
03	2	1
07	3	1

- Q.1 (a) Define Python and give features of Python.
 (b) List the standard data types in python with example.
 (c) What operators does python support?
 (d) Write the algorithm, pseudocode and draw the flowchart to find minimum in a list.

OR

What is meant by selection, iteration and sequence control structures?

07	2	1
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- Q.2 (a) What is Boolean Expression? Give an example.
 (b) What is meant by conditional If? Give the syntax of if statement.
 (c) Discuss Python Break, Continue and Pass statement.
 (d) Assuming num=125, determine the value of each of the following Python expressions.
 (i) num/125 (ii) num%100 (iii) (num==2 1) & (2<3)

02	2	2
02	2	2
03	3	2
07	3	2

OR

Write a Python program to construct the following pattern, using a nested for loop.

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*****
***
**
*
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07	3	2
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- Q.3 (a) What is len function and explain how it is used on strings with an

02	2	3
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example.

- (b) What is the use of str.upper() and str.lower() functions in string?
- (c) How to open a new file in Python?
- (d) Discuss the important functions of Python OS module.

OR

What is the use of sys module in Python.

- Q.4
- (a) What is the difference between tuples and lists in Python? 02
 - (b) Explain what is range() function and how it is used in lists? 02
 - (c) Write a few methods that are used in Python Lists. 03
 - (d) What is Dictionary? Explain Python dictionaries in detail discussing its operations and methods. 07

OR

Create a dictionary and iterate over the keys and values using for loop with index. 07

- Q.5
- (a) What is class and object in Python. 02
 - (b) What is inheritance in Python. Discuss. 02
 - (c) What is an exception? How to handle exceptions with try-except. 03
 - (d) Write a Python program to create a Vehicle class with max_speed and mileage instance attributes. 07

OR

Create a class calculator with methods of adding, subtracting, multiply, and divide. 07

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analyzing, 5-Evaluating, 6-Creating